HUST

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HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.

GENERATIVE AI IN BIOINFORMATICS, MEDICINE AND HEALTHCARE

Chapter 6: Explainable AI

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Content

6.1. Introduction to Explainable AI

6.2. Explainable AI in GenAI

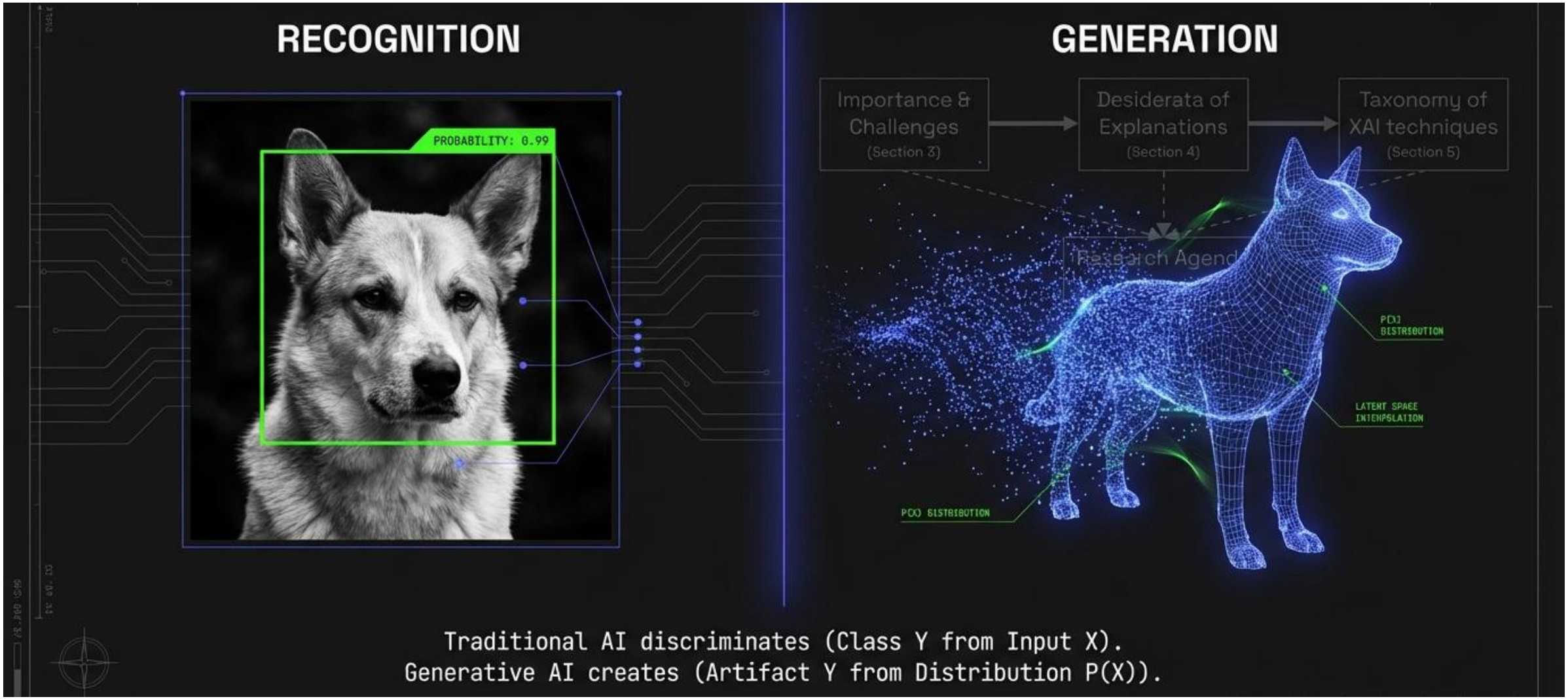
6.3. Several interpretation methods can be applied to Gen AI

6.4. Evaluate interpretation by incorporating experimental data

6.5 Identify key biological factors in AI models

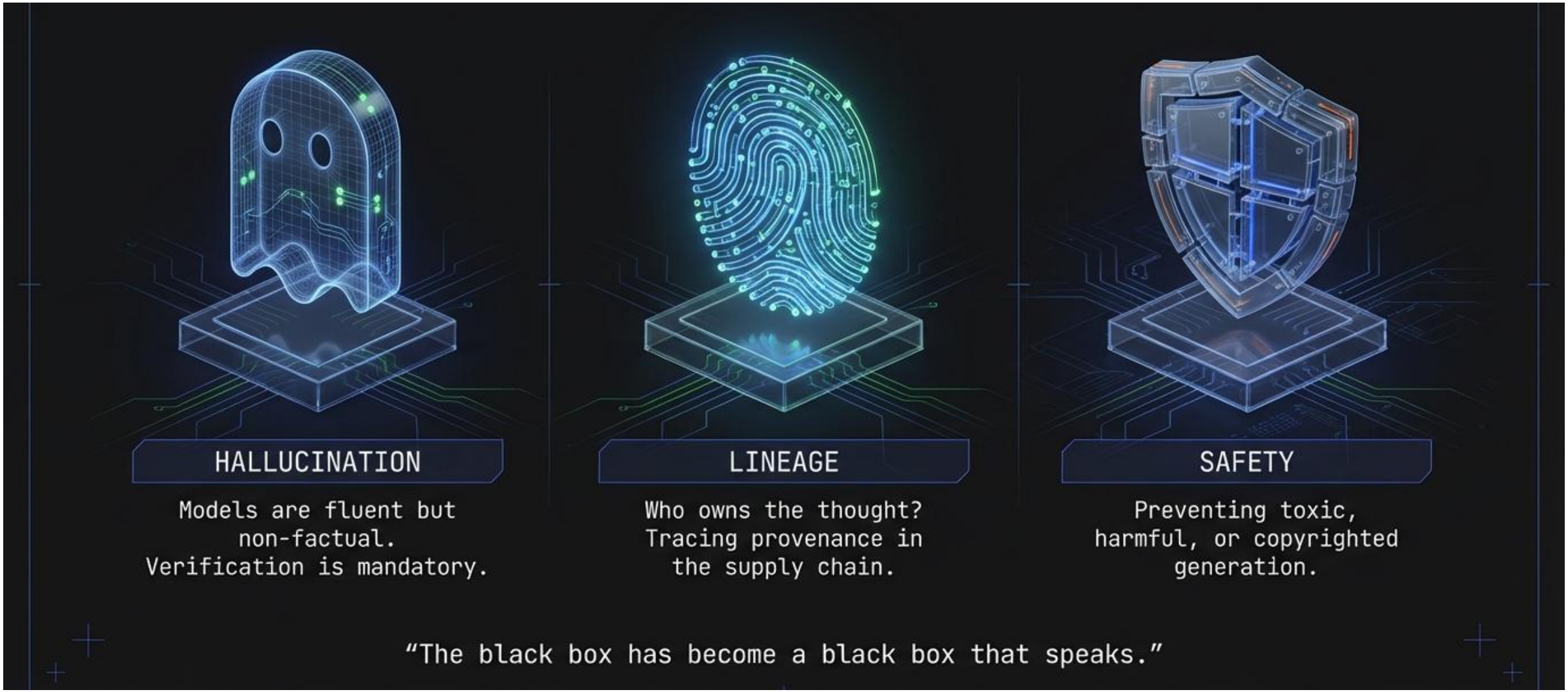
6.6. Interpretation in modern problems

The paradigm shift

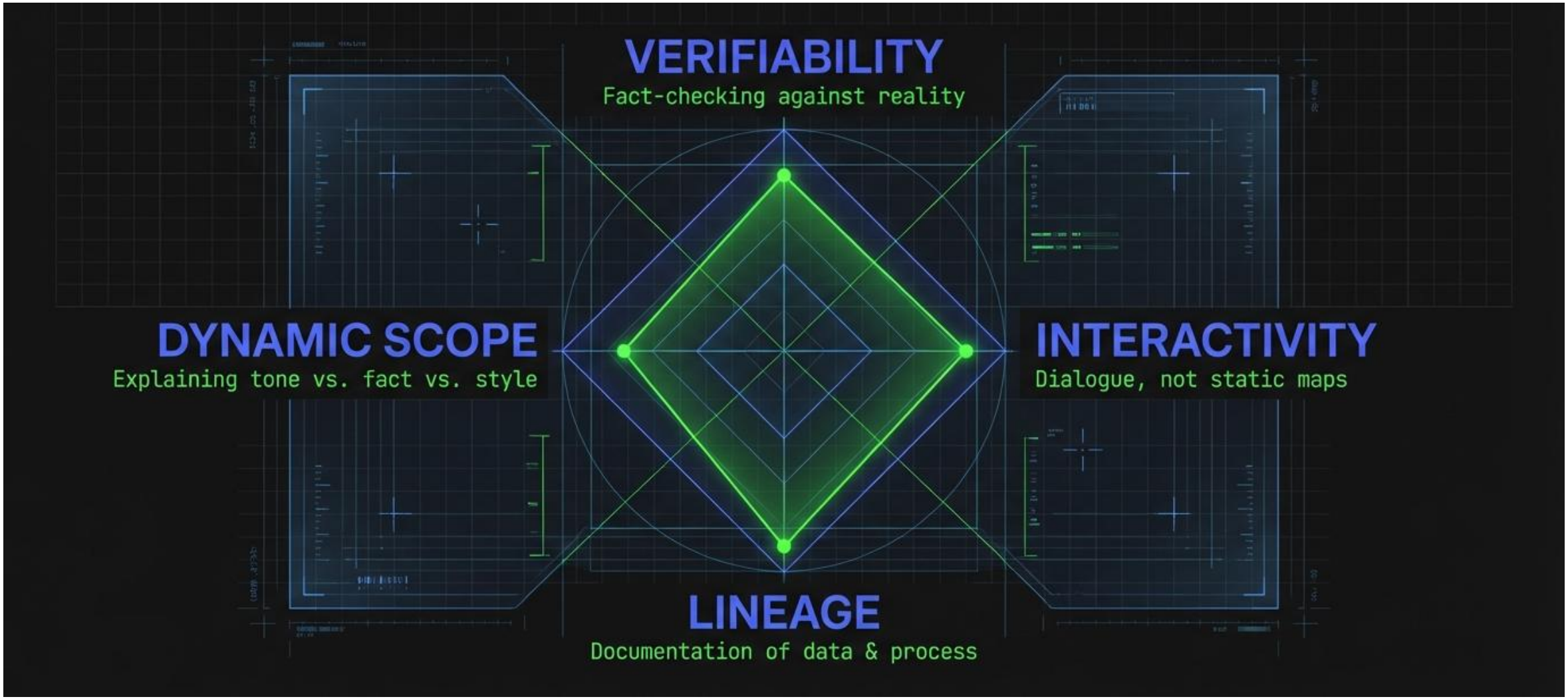


The Complexity Gap

The challenges

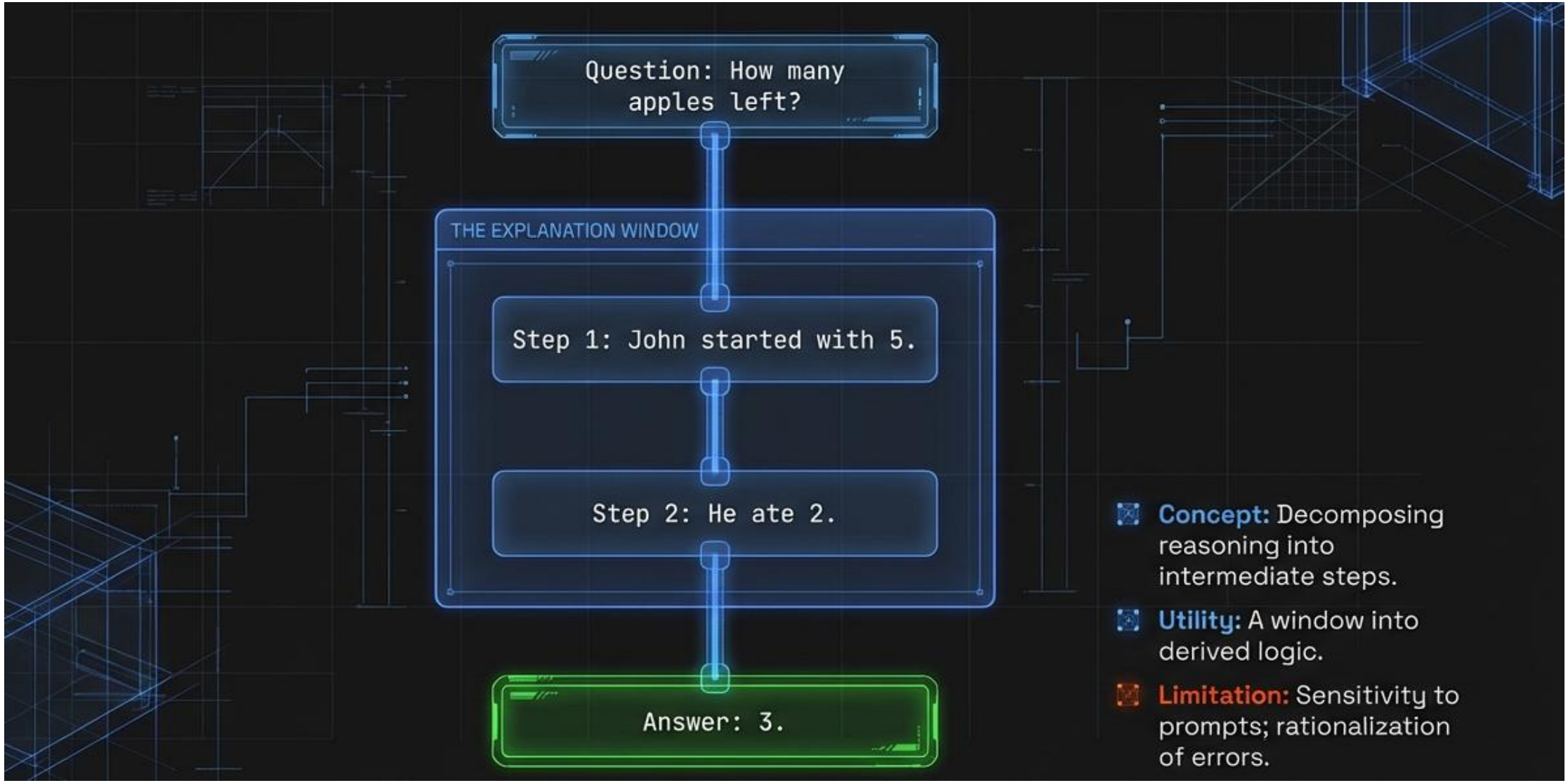


Expectation

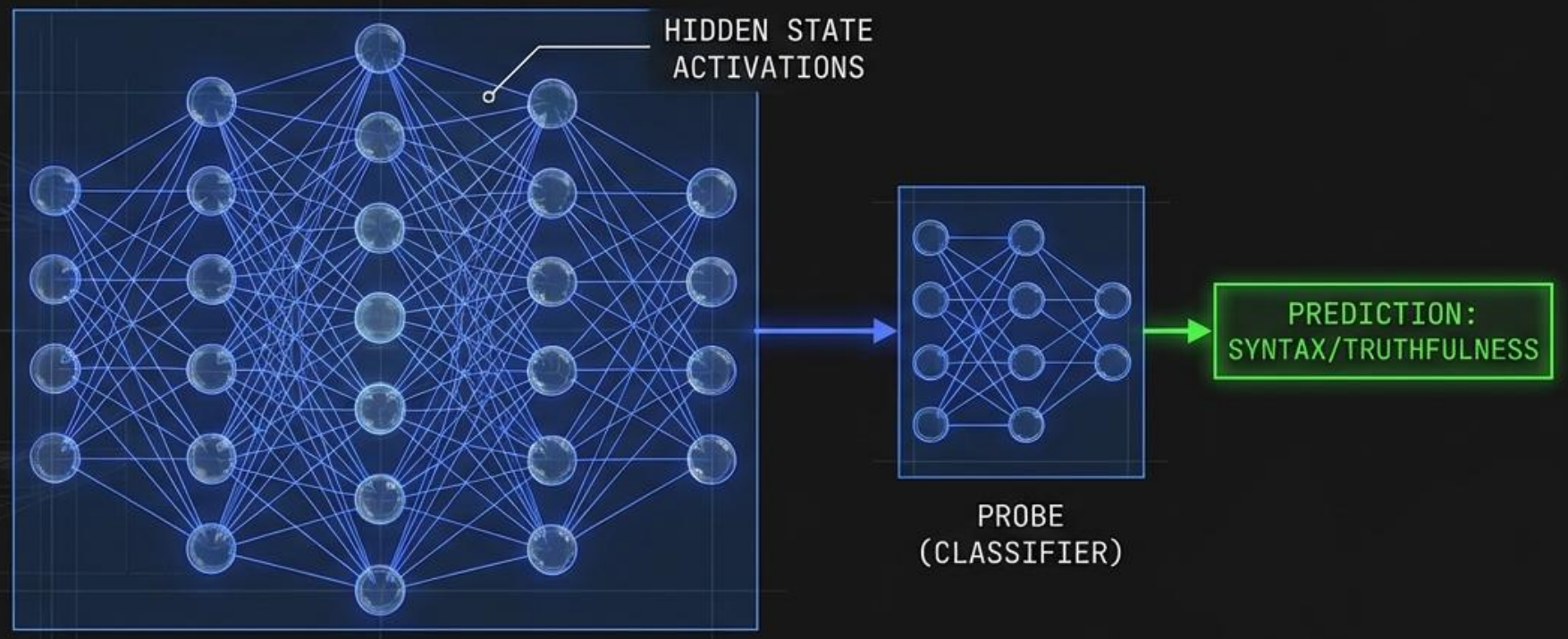


The unfaithful self-explaination

Chain-of-Thought (CoT)



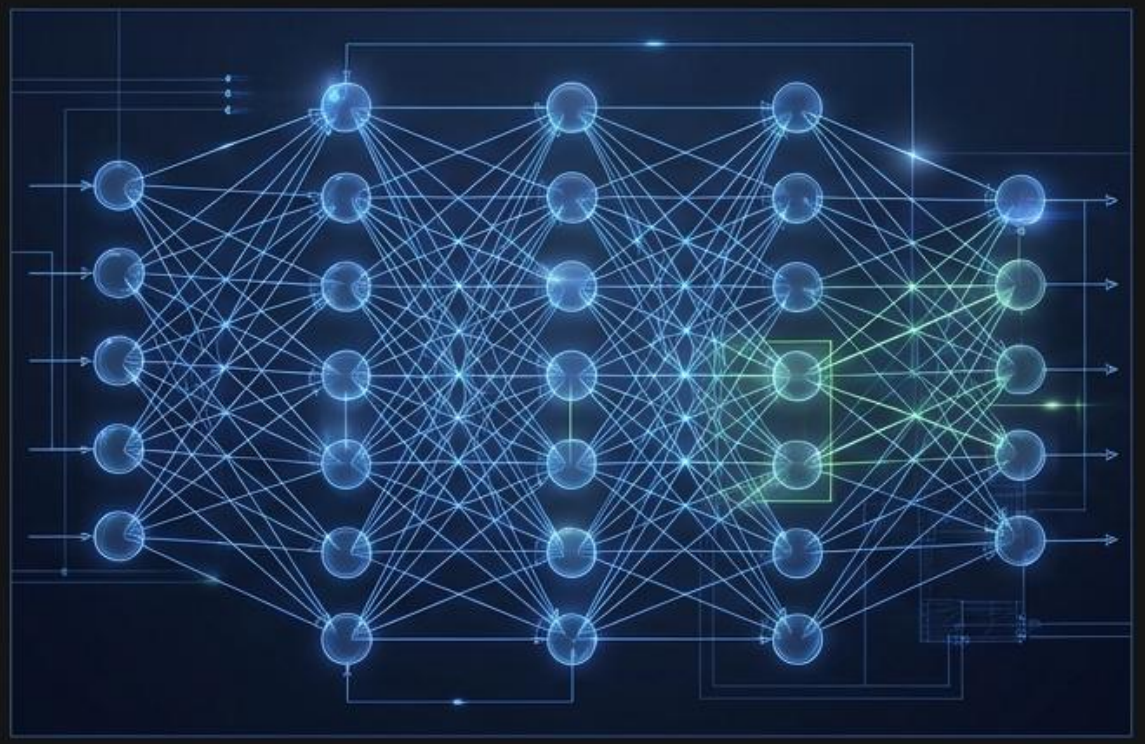
Probe the internal state

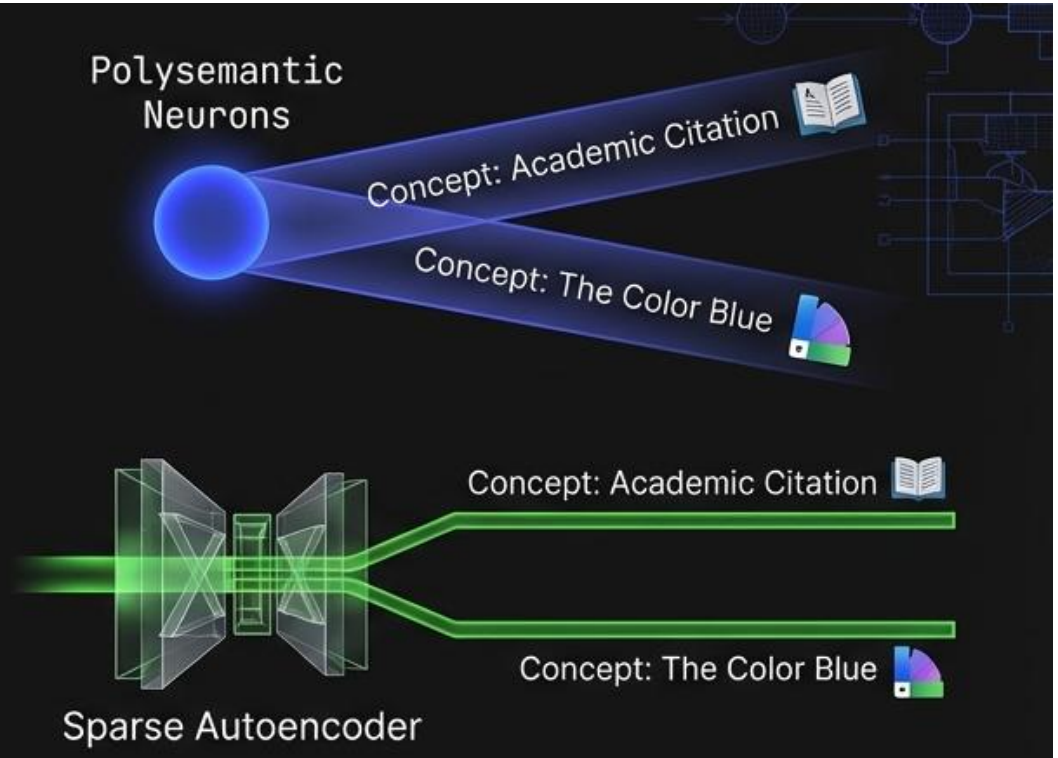


Mechanism: Training shallow classifiers on internal activations.

Goal: Locate where and when knowledge (e.g., grammar vs. fact) is acquired.

Mechanistic Intepretability





- Definition: Reverse engineering the neural circuitry.
- Problem: Polysemantic Neurons (doing too many things at once).
- Solution: Sparse Autoencoders for Dictionary Learning.

Feature Attributions in Text

[CLS] has a lot of the virtues of eastwood at his best.

Question: Which input word caused the model to generate 'virtues'?

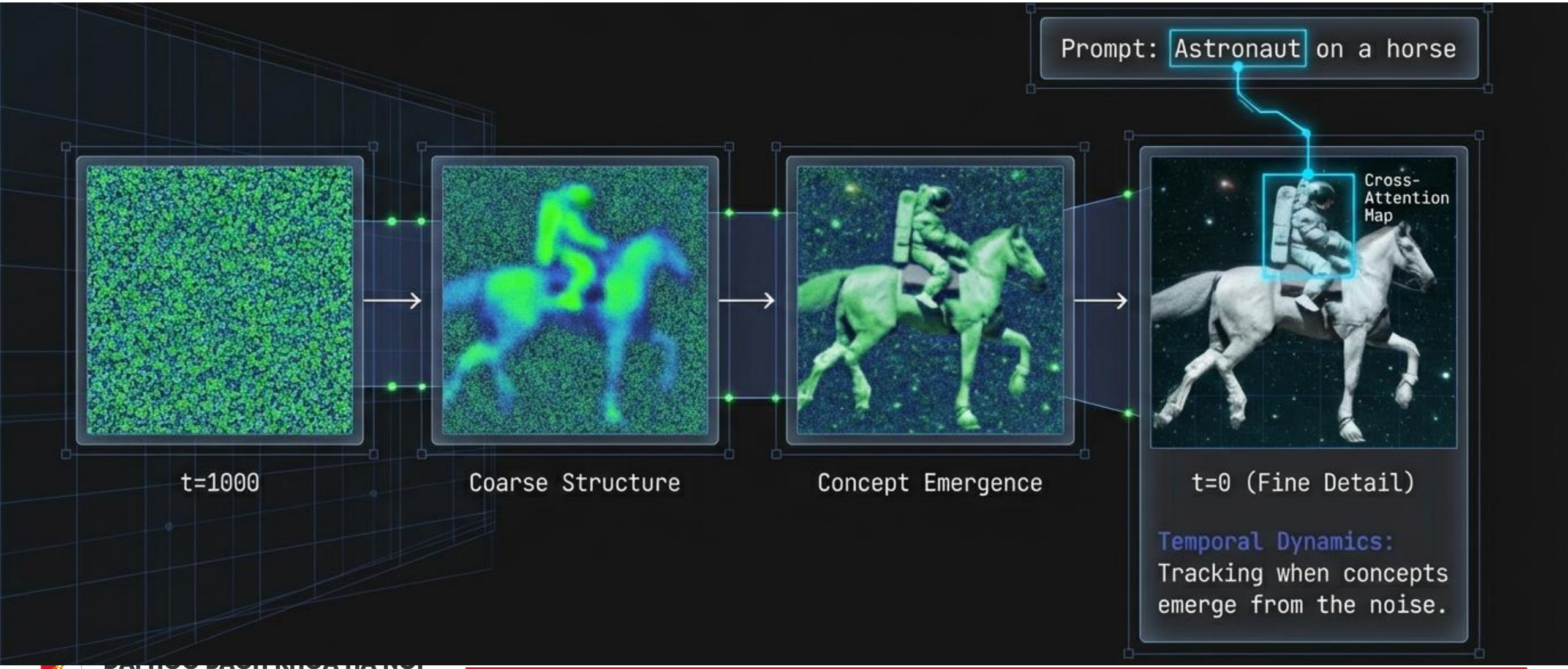
- Method: Adapting SHAP/LIME for token generation.
- Constraint: Computationally expensive for long sequences.

Visualization Imagination

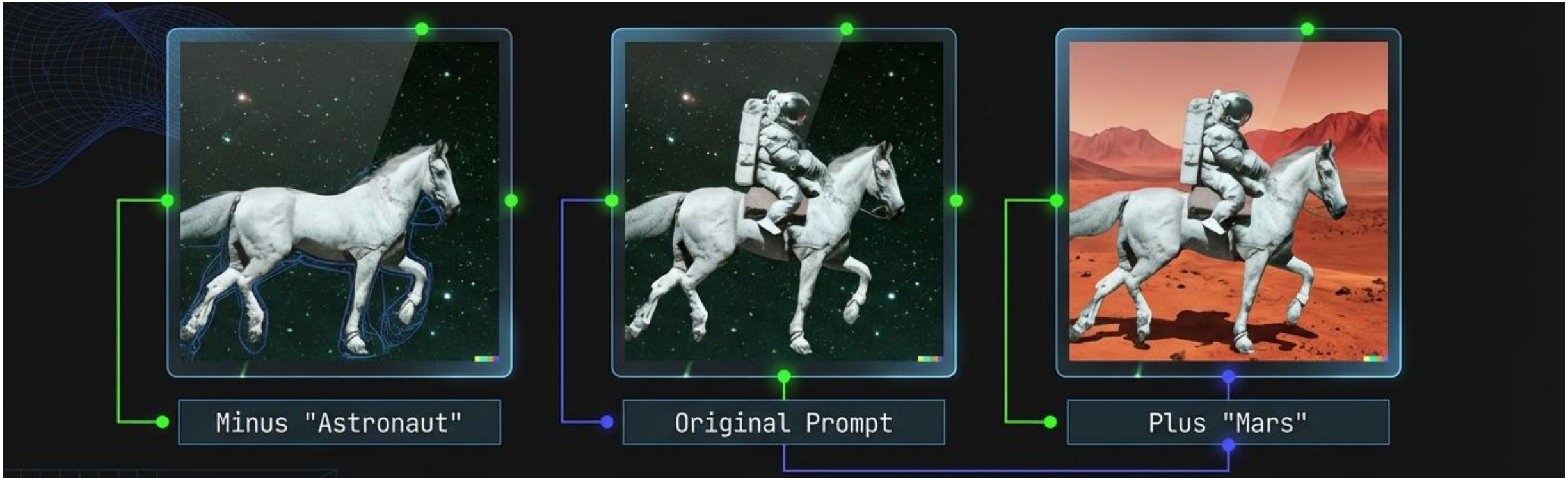
Distanglement in gains

- **Concept:** Separating latent variables (z-vector) to control generative factors.
- **Goal:** Verifying the model has learned distinct concepts, not just memorization.

Inside the diffution process

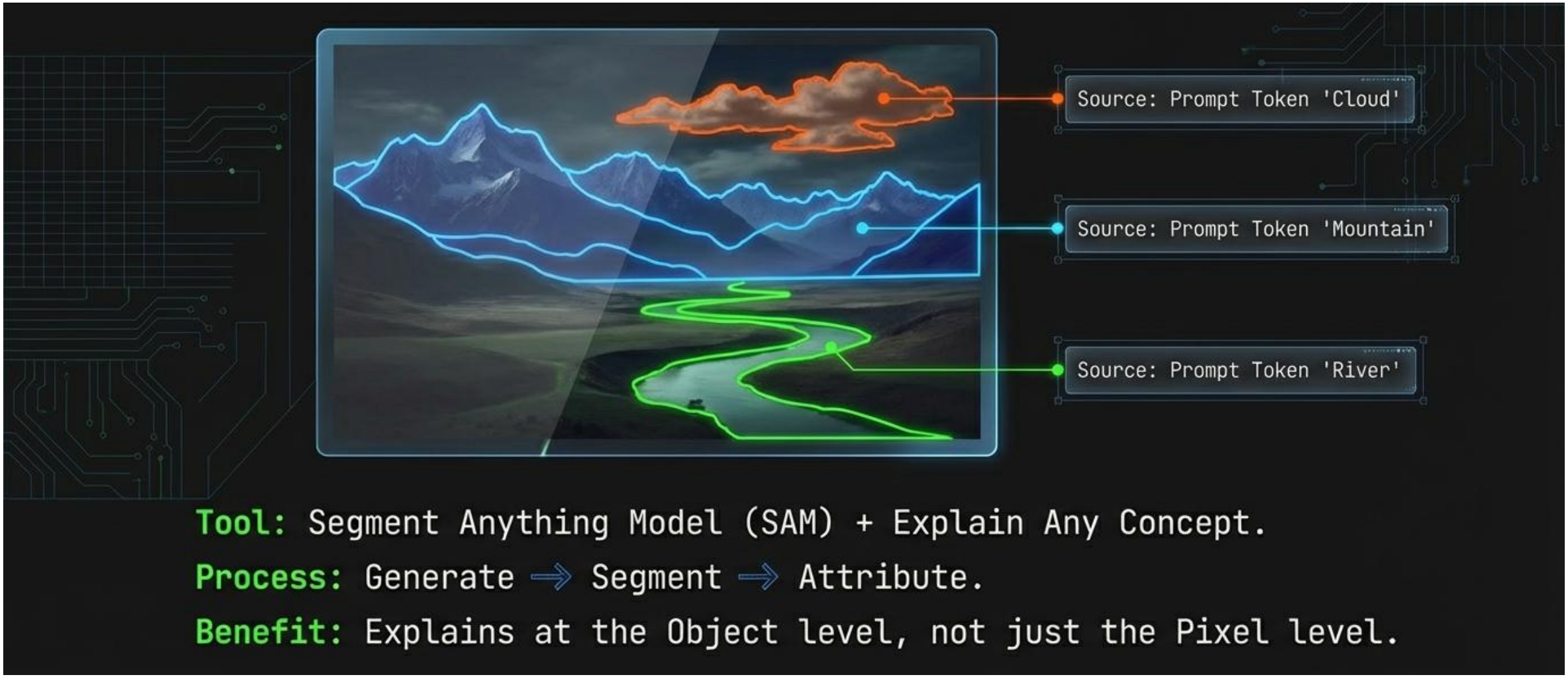


Counterfactual explaination

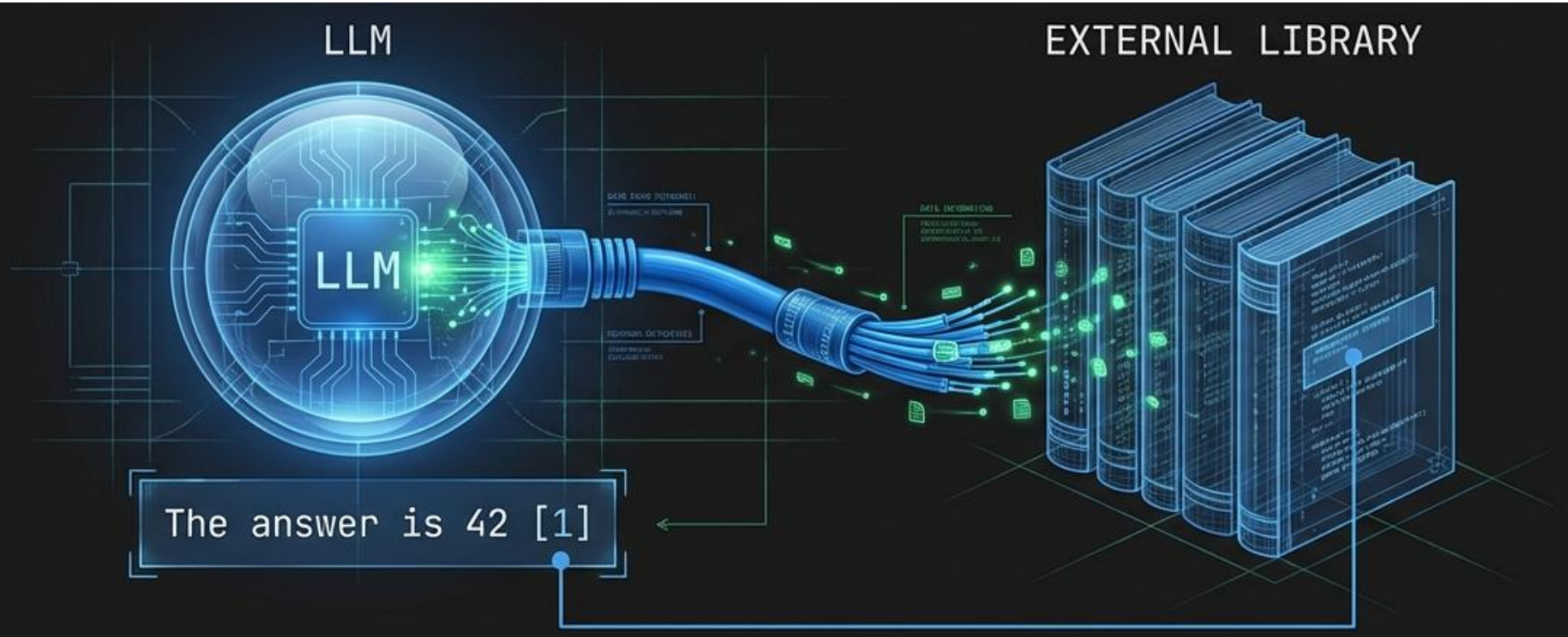


- **Method:** Diffusion Models for Counterfactual Explanations (DiME).
- **Question:** "If I remove X, how does the image change?"
- **Insight:** Checks for causal dependence and entanglement.

Concetp Attribution



Retrievel Augumented Generation as XAI



- **Mechanism:** The "Open Book" exam. Retrieving documents before generating.
- **Explanation:** Citing sources directly.
- **Verifiability:** High. Users can check the source.

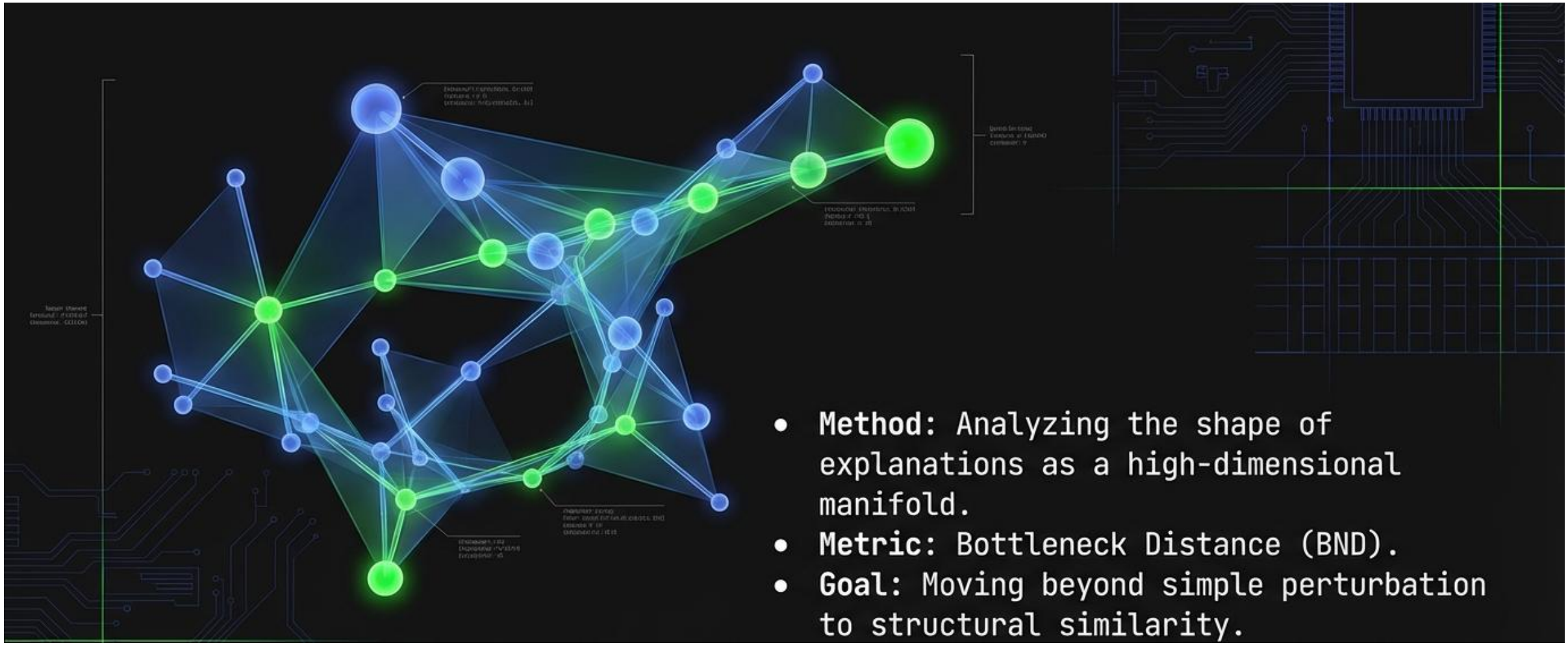
The disagreement problem

		synthetic			german			adult			spambase			har		
spearmanr	ABC	1.0	0.38	0.59	1.0	0.21	0.0	1.0	0.26	-0.029	1.0	0.73	0.24	1.0	0.077	0.68
	BND	0.38	1.0	0.55	0.21	1.0	-0.21	0.26	1.0	-0.087	0.73	1.0	-0.011	0.077	1.0	0.042
	PGI	0.59	0.59	1.0	0.0	-0.21	1.0	-0.029	-0.087	1.0	0.24	-0.011	1.0	0.68	0.042	1.0
wtd tau	ABC	1.0	0.44	0.5	1.0	-0.079	-0.031	1.0	0.21	-0.091	1.0	0.7	0.086	1.0	0.094	0.45
	BND	0.44	1.0	0.55	-0.079	1.0	-0.19	0.21	1.0	0.15	0.7	1.0	0.19	0.094	1.0	-0.048
	PGI	0.5	0.55	1.0	-0.031	-0.33	1.0	-0.091	0.15	1.0	0.086	0.19	1.0	0.45	-0.048	1.0
kendall tau	ABC	1.0	0.28	0.44	1.0	0.12	0.044	1.0	0.39	0.033	1.0	0.56	0.16	1.0	0.022	0.49
	BND	0.28	1.0	0.41	0.12	1.0	-0.088	0.19	1.0	-0.057	0.59	1.0	0.0	0.022	1.0	0.022
	PGI	0.44	0.41	1.0	0.044	-0.088	1.0	0.033	-0.057	1.0	0.16	0.0	1.0	0.49	0.022	1.0
		ABC	BND	PGI	ABC	BND	PGI	ABC	BND	PGI	ABC	BND	PGI	ABC	BND	PGI

Crisis: Metrics like PGI, PGU, and Ablation disagree on what counts as 'faithful'.

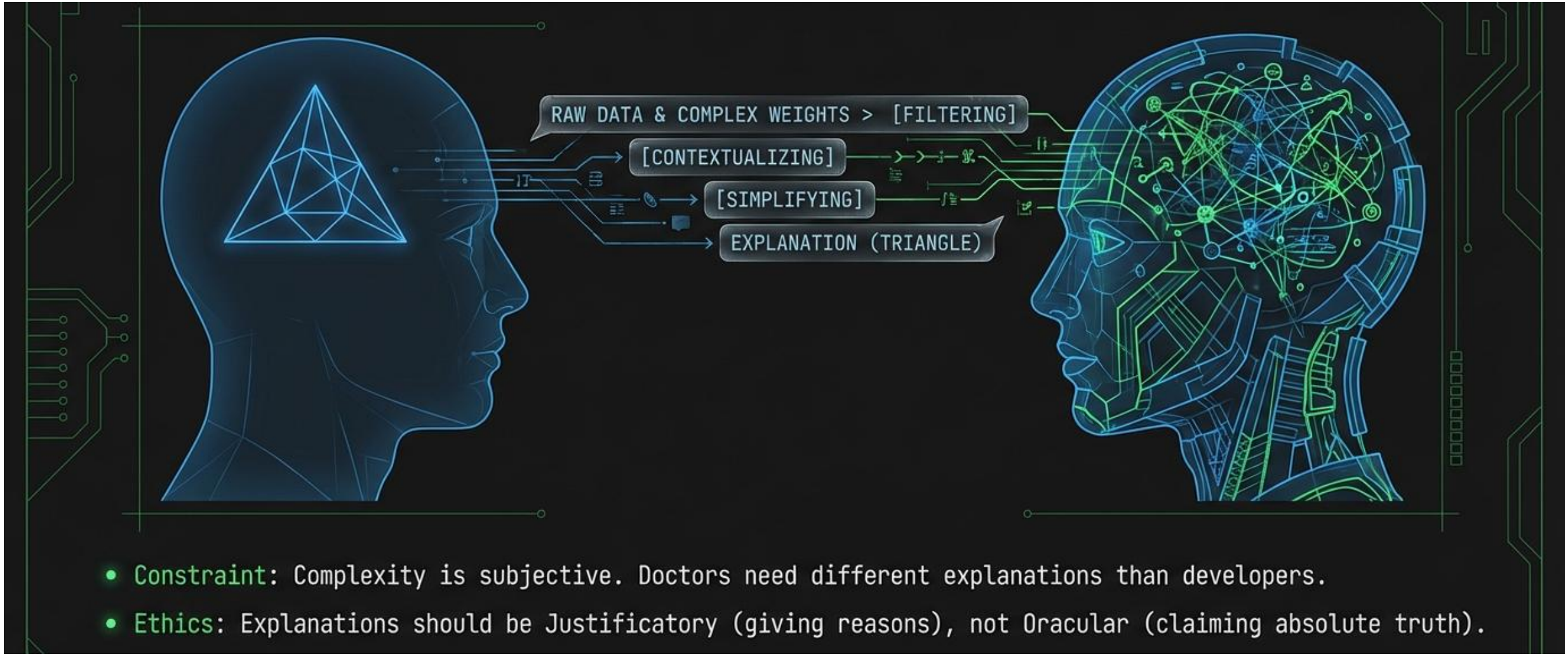
Implication: If metrics disagree, we don't know which explanation to trust.

Topological Evaluation



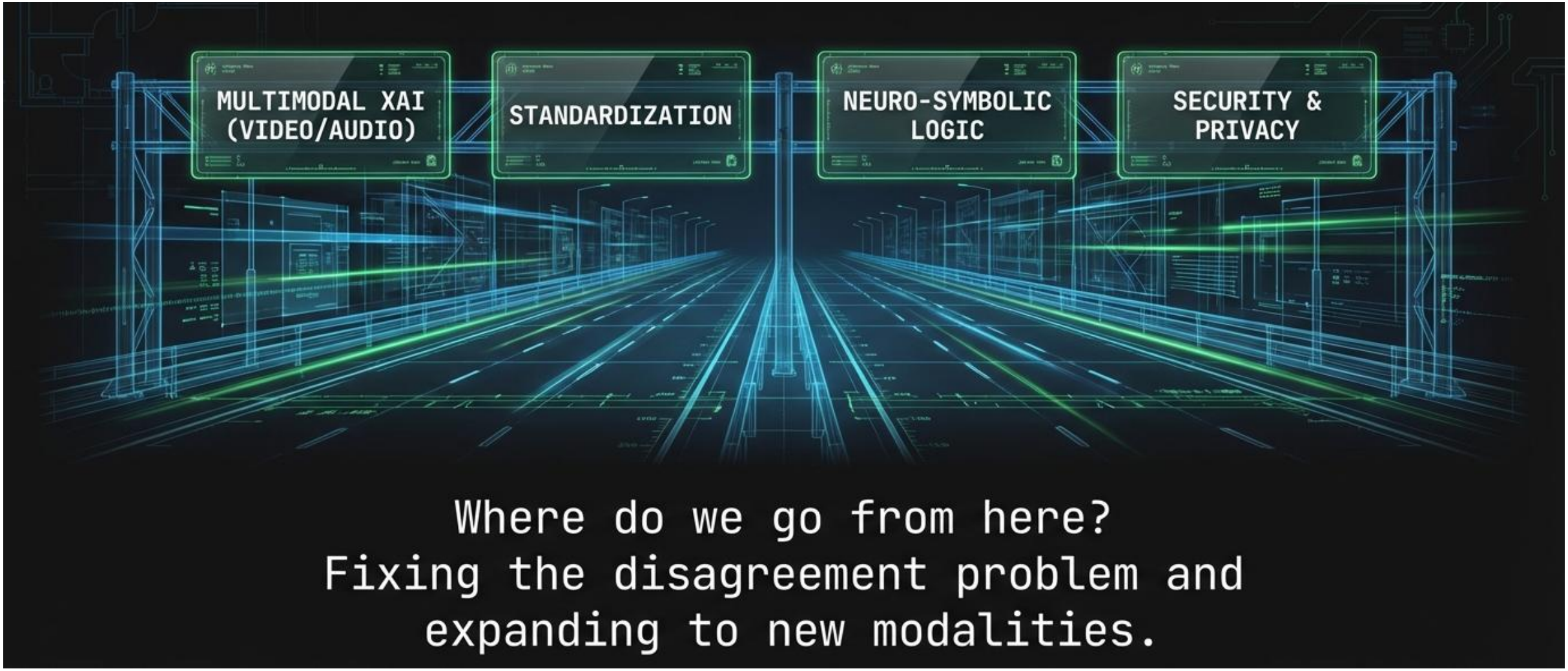
- **Method:** Analyzing the shape of explanations as a high-dimensional manifold.
- **Metric:** Bottleneck Distance (BND).
- **Goal:** Moving beyond simple perturbation to structural similarity.

Human-centric XAI

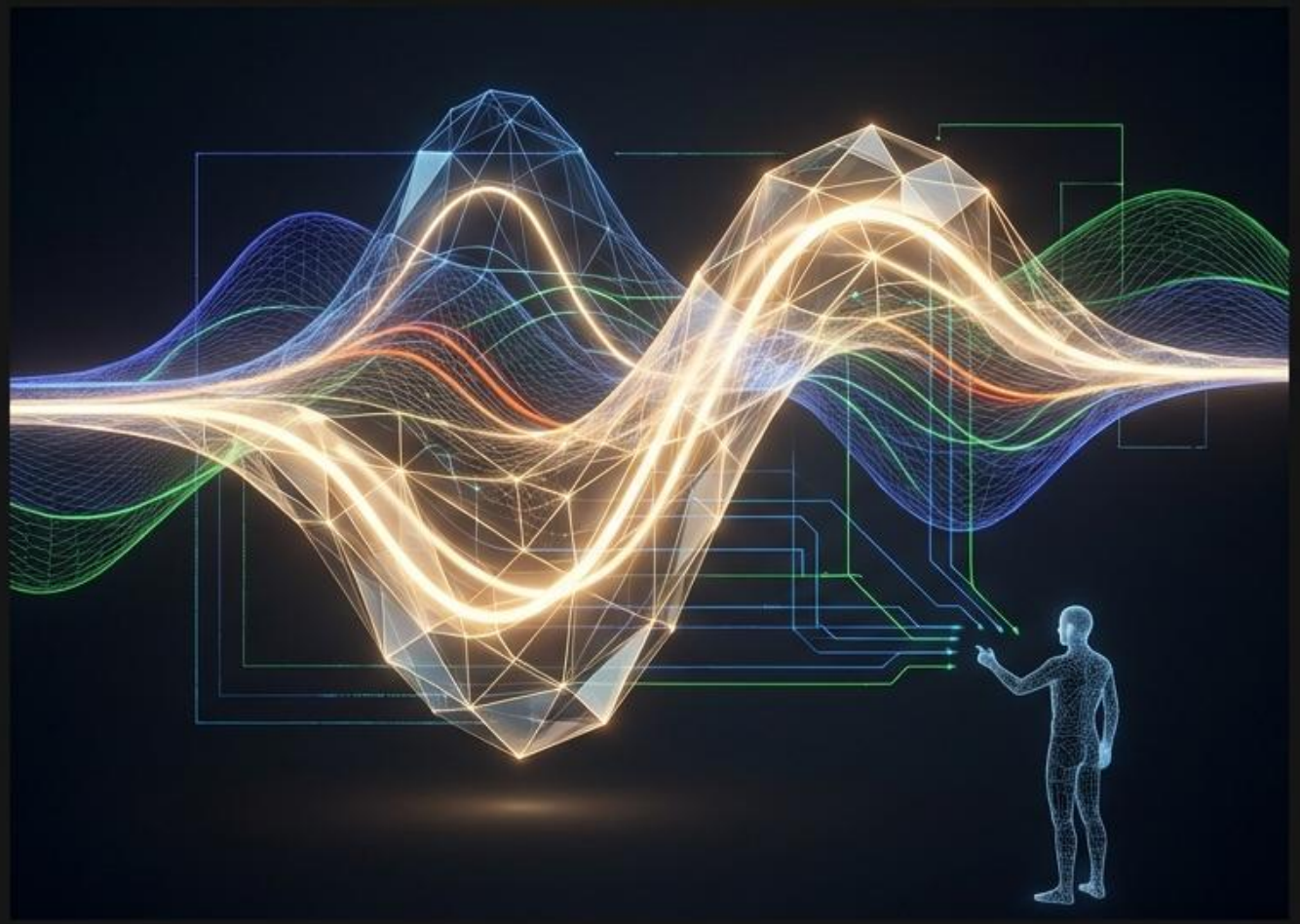


- Constraint: Complexity is subjective. Doctors need different explanations than developers.
- Ethics: Explanations should be Justificatory (giving reasons), not Oracular (claiming absolute truth).

Research Agenda



Conclusion



SUMMARY: From static heatmaps to dynamic, interactive dialogue.

FINAL THOUGHT: XAI is the user interface for the future of Generative AI.

Based on research by Schneider et al., Longo et al., Barr et al., and Zhou et al.

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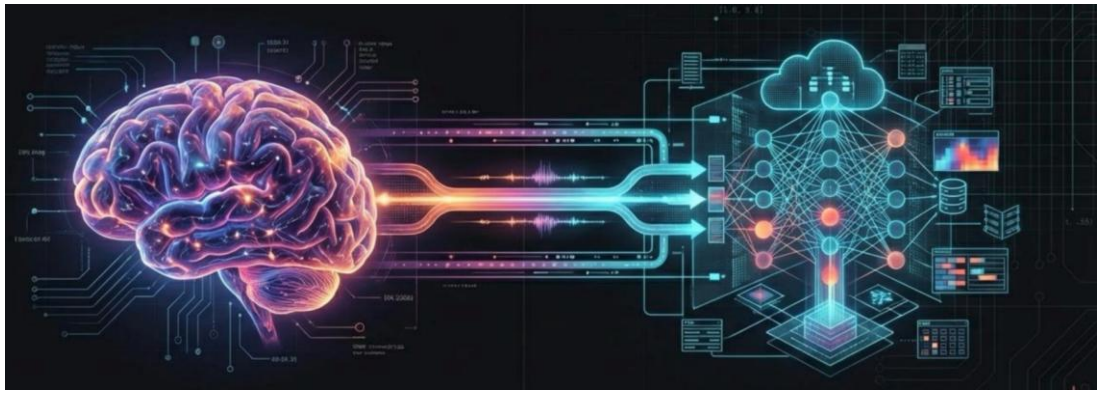
Thanks